Hat CBack NBIoT

SKU:A113



















Description

Hat CBack NBIoT is a highly integrated NB-IoT communication module compatible with the M5StickC/C Plus series. It features a built-in SIM7020G module and a SIM-CARD slot (nano slot), covering Cat-NB frequency bands in most regions worldwide. The antenna is connected via an SMA connector and comes with a rubber duck antenna. This module supports various common IoT communication protocols (MQTT, HTTP, CoAP, etc.), making it suitable for applications in low-latency, low-throughput scenarios.

Features

- Compatible with M5StickC/C Plus series
- AT command control, UART transmission rate: 115200bps
- SIM card type: Nano
- External antenna: SMA antenna
- SIM7020G certified in multiple countries

Includes

- 1 x Hat CBack NBIoT
- 1 x SMA antenna
- 2 x M2x5 screws

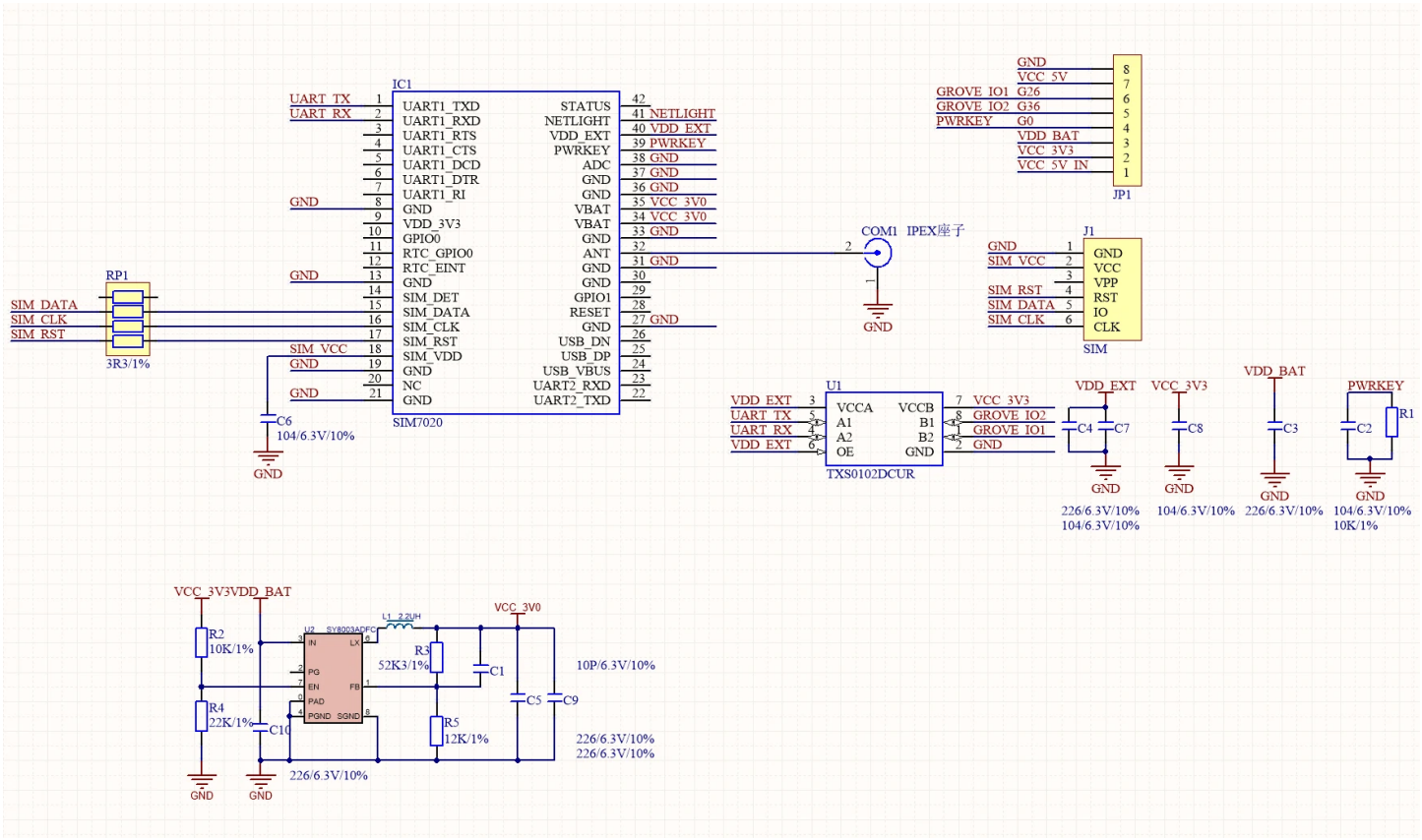
Applications

- Smart metering
- Remote monitoring
- Bike sharing

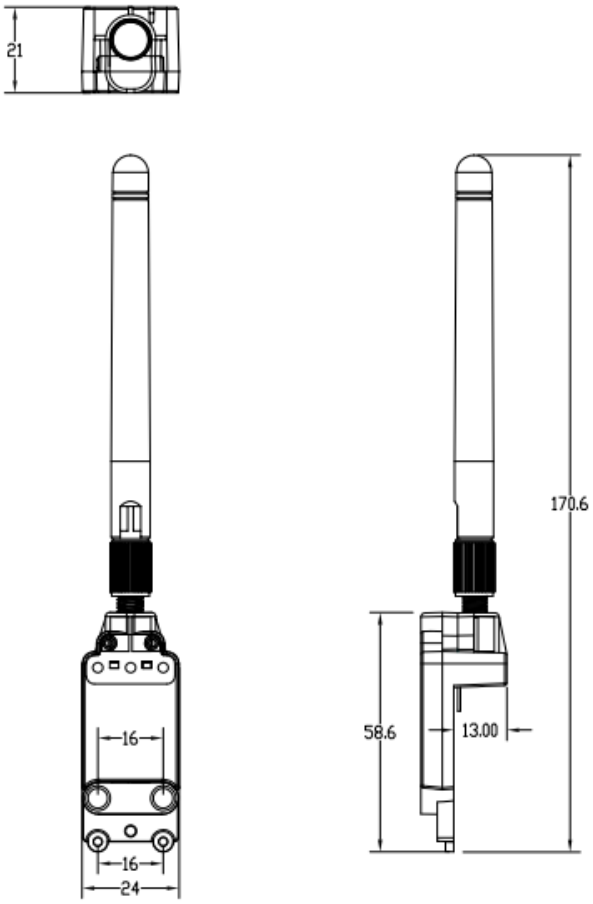
Specifications

Specification	Parameters
Communication Module	SIM7020G
Supported Cat-NB Bands	B1/B2/B3/B4/B5/B8/B12/B13/B17/B18/B19/B20/B25/B26/B28/B66/B70/B71/B85
Data Transfer Speed	126(DL)/150(UL)
Network Protocols	TCP/UDP/LWM2M/COAP/ MQTT/HTTP/HTTPS/ TLS/DTLS/DNS/NTP/ PING/OneNET/Telecom Cloud/Aila Cloud*
Communication Method	UART 115200bps
Certifications	CE*/FCC*/GCF*/PTCRB*/RoHS/REACH/ATEX/TIM*/Deuts Telecom*/Vodafone*/T-Mobile*
Product Size	68 × 24 × 21mm
Package Size	90 × 35 × 21mm
Product Weight	12.2g
Gross Weight	25g

Schematics


IC1
UART_TX
UART_RX
GND
SIM_VCC
GND
GND
UART1_TXD
UART1_RXD
UART1_RTS
UART1_CTS
UART1_DCD
UART1_DTR
UART1_RI
GND
VDD_3V3
GPIO0
RTC_GPIO0
RTC_EINT
GND
SIM_DET
SIM_DATA
SIM_CLK
SIM_RST
SIM_VDD
GND
NC
GND
STATUS
NETLIGHT
VDD_EXT
PWRKEY
ADC
GND
GND
VBAT
VBAT
GND
ANT
GND
GND
GPIO1
RESET
GND
USB_DN
USB_DP
USB_VBUS
UART2_RXD
UART2_TXD
SIM7020
RP1
SIM_DATA
SIM_CLK
SIM_RST
3R3/1%
C6
104/6.3V/10%
GND
COM1 IPEX座子
GND
GND
VCC_5V
GROVE_IO1 G26
GROVE_IO2 G36
PWRKEY G0
VDD_BAT
VCC_3V3
VCC_5V_IN
JP1
J1
GND
SIM_VCC
SIM_RST
SIM_DATA
SIM_CLK
GND
VCC
VPP
RST
IO
CLK
SIM
U1
VDD_EXT
UART_TX
UART_RX
VDD_EXT
VCCA
A1
A2
OE
VCCB
B1
B2
GND
VCC_3V3
GROVE_IO2
GROVE_IO1
GND
TXS0102DCUR
VDD_EXT
C4
C7
GND
226/6.3V/10%
104/6.3V/10%
VCC_3V3
C8
GND
104/6.3V/10%
VDD_BAT
C3
GND
226/6.3V/10%
PWRKEY
C2
R1
GND
104/6.3V/10%
10K/1%
VCC_3V3VDD_BAT
R2
10K/1%
R4
22K/1%
C10
GND
GND
U2 SY8003ADFC
IN
LX
PG
EN
FB
PAD
PGND SGND
L1 2.2UH
R3
52K3/1%
R5
12K/1%
226/6.3V/10%
VCC_3V0
C1
10P/6.3V/10%
C5
C9
226/6.3V/10%
226/6.3V/10%

Model Size



Datasheets

- SIM7020 Series datasheet
- SIM7020 Series_Ayla_Application Note_V1.03
- SIM7020 Series_CTBURST_Application Note_V1.01
- SIM7020 Series_CoAP_Application Note_V1.03
- SIM7020 Series_EAT_Environment & Compilation & Burning Guide_V1.02.
- SIM7020 Series_FOTA_Application_Note_V1.02
- SIM7020 Series_HTTP(S)_Application Note_V1.04
- SIM7020 Series_LWM2M_Application Note_V1.03
- SIM7020 Series_Low Power Mode_Application Note_V1.05
- SIM7020 Series_MQTT(S)_Application Note_V1.05
- SIM7020 Series_NVRAM_Application Note_V1.02
- SIM7020 Series_SAT_Application Note_V1.01
- SIM7020 Series_SNTP_Application Note_V1.03
- SIM7020 Series_TCPIP_Application Note_V1.04
- SIM7020 Series_AT Command Manual_V1.05

Softwares

Arduino

- CBack NBIoT-CN MQTT Example

- CBack NBIoT-CN HTTP Example

UiFlow1

- CBack NBIoT-CN UiFlow1 Docs